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10 August 1983
SUPERSEDING
MS 14418
2 August 1973

MILITARY SPECIFICATION

DIBUTYLTIN DILAURATE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the minimum requirements for one grade of dibutyltin dilaurate.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specification and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARD

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1218	ACS Chemicals

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connections with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (524), Indian Head, MD 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

(ASTM)

D 1544-80	Color of Transparent Liquids (Gardner Color Scale)
D 1963-74	Specific Gravity of Drying Oils, Varnishes, Resins, and Related Materials at 25/25° C

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

VARIAN ASSOCIATES, PALO ALTO, CALIFORNIA

Analytical Methods for Flame Spectroscopy

(Application for copies should be addressed to Varian Associates, 611 Hansen Way, Palo Alto, CA 94303.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENT

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.4).

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3.2 Chemical and physical properties. The dibutyltin dilaurate shall meet the requirements of Table I.

TABLE I. Chemical and physical properties.

Property	Requirement
Tin content,%	17.7 - 19.5
Color, Gardner	4 max.
Acid number	130 - 180
Specific gravity at 25°C	1.045 - 1.054

3.3 Workmanship. The material shall be uniform in quality and free from foreign materials that could adversely affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), inspection conditions shall be as specified in the applicable test method.

4.4 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on material which has been produced from raw materials with equipment and procedures normally used in production.

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When specified in the conract (see 6.2), the contractor shall prepare a report giving the results obtained for all inspections and tests performed. Unless otherwise specified, all chemicals shall be ACS grade in accordance with MIL-STD-1218 and distilled water shall be used. Where applicable, blank determinations shall be run and corrections applied where significant. First article approval is valid on the contract under which it is granted, unless extended by the Government to other contracts.

4.4.1 Sample size. Sample size shall be as specified in the contract (see 6.2).

4.4.2 Inspection routine. The sample shall be subject to the tests of 4.6 plus any additional tests specified by the procuring activity (see 6.2).

4.4.3 Failures. Failure of the sample to pass any test shall be cause for first acticle rejections.

4.5 Quality conformance inspection. The material shall be subject to all the following inspections and tests for acceptance. When specified in the contract (see 6.2), the contractor shall prepare a report giving the results obtained for all inspections and tests performed and a certified statement that the lot meets all the requirements of this specification. Unless otherwise specified, all chemicals shall be ACS grade in accordance with MIL-STD-1218 and distilled water shall be used. Where applicable, blank determinations shall be run and corrections applied where significant.

4.5.1 Sampling.

4.5.1.1 Lot. For the purpose of sampling, a lot shall consist of material of a single form manufactured under one set of conditions from one production batch and offered for delivery at one time.

4.5.1.2 Sampling for examination of product. A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105, inspection level II.

4.5.1.3 Sampling for tests. A representative sample of sufficient material to form the required tests shall be taken from each lot.

4.6 Test methods. Tests shall be performed in accordance with the following methods.

4.6.1 Tin content. The wet analytical method of 4.6.1.1 or the alternate instrumental method of 4.6.1.2 may be used.

4.6.1.1 Titrimetic method.

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4.6.1.1.1 Apparatus and reagents.

a. Erlenmeyer flasks, 500 millilters (ml)

b. Hot plate

c. Meeker burner

d. Concentrated sulfuric acid, c.p. reagent

e. Fuming nitric acid, c.p. reagent

f. Hydrochloric acid, 2.1 (this acid should be boiled to remove air and placed in a stoppered bottle for future analyses)

g. Standard 0.1N potassium iodate (prepared from water previously boiled)

h. Potassium iodide, reagent 10 percent solution (prepared from water previously boiled)

i. Water, boiled to remove all air

j. Starch solution (prepared from previously boiled water)

k. Iron powder, 99 percent Fe hydrogen reduced

l. Nickel coils. Roll into loose coil a 6- by 3- by 0.015-inch piece of nickel sheet. Before using the first time, clean in petroleum ether and then place in HC1 (2:1) and boil for about 5 minutes. Remove and rinse with H_2O.

m. Perchloric acid (optional, but preferred).

4.6.1.1.2 Procedure.

4.6.1.1.2.1 Preparation of sample.

a. Weigh 0.4- to 0.5-gram (g) sample accurately to the nearest milligram (mg) via a 1-ml syringe into a 500-ml Erlenmeyer flask. Add 50 ml of fuming nitric acid. Swirl.

b. Add 20 ml of sulfuric acid. Swirl and add a few glass beads and 1 ml of perchloric acid.

c. Place on hot plate set at high heat; allow to boil until dense white fumes of sulfur trioxide appear. Fume for 10 minutes.

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d. Using crucible tongs swirl the flask over a Meeker burner and fume for 1 to 2 minutes. Solution should be colorless at this point. If not, add fuming nitric acid dropwise and bring to dense white fumes on hot plate. Repeat Meeker burner step.

e. Cool; wash down the sides of the flask with 30 ml of water. Place on hot place and evaporate the solution to heavy fumes of sulfur trioxide.

f. Cool and repeat step "e".

g. Cool, add 80 ml of water and 100 ml of 2:1 hydrochloric acid and 100 ml of water in that order.

4.6.1.1.2.2 Reduction and titration of tin.

a. To the prepared solution, add a nickel coil and 5 g of iron powder. Swirl.

b. Wash down the sides of the flask with water.

c. Boil gently for about 20 minutes.

d. Cool. Add a few pieces of Dry Ice.

e. Add 5 ml of starch solution, 5 ml of 10 percent potassium iodide.

f. Titrate with potassium iodate to starch-iodine endpoint.

g. Calculation:

$$\% \text{ Tin} = \frac{\text{Vol. X N X } 5.89}{\text{Wt. of sample}}$$

Where:

Vol. = milliliters of potassium iodate used

N = normality of potassium iodate.

4.6.1.2 Instrumental method, atomic absorption.

4.6.1.2.1 Apparatus. An atomic absorption spectrophotometer such as Varian Techtron AA-5 or Perkin-Elmer Model 303 equipped with an acetylene and nitrous oxide gas system or any equivalent atomic abosrption spectrophotometer.

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4.6.1.2.2 Materials. ACS grade xylenes, dibutyltin bis (2-ethylhexanoate) (Eastman 10427 for atomic absorption standards), atomic absorption lamp for tin.

4.6.1.2.3 Procedure. Method shall be that described for "Tin in Organometallic Salts" in Varian Associates Manual "Analytical Methods for Flame Spectroscopy." Sample weight shall be 0.6 g of dibutyltin dilaurate weighed accurately to 0.1 (mg). Final sample dilution shall represent 0.600 g per 1000 ml. Wavelength shall be 286.3 nano-meters. Standards and sample dilution may be altered to better suit capabilities of a particular instrument with absorbency of the solution falling between 0.2 and 1.0.

4.6.2 Gardner color. Gardner color shall be determined in accordance with ASTM D 1544.

4.6.3 Acid number. Weigh a 4- to 5g-sample to the nearest mg on an analytical balance. Dissolve in 250 ml of isopropanol and 50 ml of water. (If a haze develops add an additional 25 ml of isopropanol.) Titrate to a phenolphthalein endpoint using 1.0N KOH (pink color for at least 30 seconds). Calculate acid number as follows:

$$\text{Acid number} = \frac{\text{ml KOH X normality X 56.1}}{\text{wt. of sample}}$$

4.6.4 Specific gravity. Specific gravity shall be determined in accordance with ASTM D 1963.

4.6.5 Examination of product. The samples taken in accordance with 4.5.1.2 shall be inspected to verify compliance with the requirements of section 5 and other requirements of this specification not involving tests.

5. PACKAGING

5.1.1 Commercial. Unless otherwise specified, the material shall be packaged in accordance with the manufacturer's best accepted commercial practice.

5.2 Packing

5.2.1 Commercial. Unless otherwise specified, the material shall be packed in shipping containers in such a manner as to insure safe transportation by common carrier. Containers used shall comply with National Motor Freight Classification Rules and the Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with the provisions of MIL-STD-129.

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6. NOTES

6.1 Intended use. The dibutyltin dilaurate covered by this specification is intended for use in the manufacturer of liner material for torpedo warheads and plastic bonded explosives.

6.2 Ordering data. Acquisition documents should specify the following:

6.2.1 Acquisition requirements.

a. Title, number, and date of this specification
b. Quantity desired
c. Inspection conditions, if other than as specified (see 4.3)
d. First article sample size, if required (see 4.4.1)
e. Additional first article tests, if any (see 4.4.2)
f. First article instructions regarding arrangements for examinations, test and approval
g. Request for tests results and certificate (see 4.4, 4.5 and 6.2.2)
h. Any special marking required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.4	Test report	DI-T-2072
4.5	Certification data/report	UDI-A-23264

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

Preparing activity:
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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